



10562 - Spectroscopy properties of young massive clusters in the Antennae: How common are very massive stars at high metallicity?

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Paul A. Crowther (PI) (ESA Member)	University of Sheffield
Dr. Linda J. Smith (CoI) (US Admin CoI)	Space Telescope Science Institute
Dr. Calum Hawcroft (CoI)	Space Telescope Science Institute
Dr. Peter Senchyna (CoI)	Carnegie Institution of Washington
Prof. John Chisholm (CoI)	University of Texas at Austin
Dr. Iker Millan Irigoyen (CoI) (ESA Member)	Max Planck Institut fur Aeronomie

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	9	PA81-4configs-G140H-G235H-G395H-midpoint	NIRSpec MultiObject Spectroscopy	(7) YMC_Antennae

ABSTRACT

Historically the upper stellar mass limit of 100 Msun was widely accepted, but increasing evidence from both resolved and unresolved stellar populations favours the existence of very massive stars (VMS). We propose NIRSpec/MSA observations of 28 young massive clusters (YMCs) within the merging Antennae (NGC 4038/9) galaxies, a sample size sufficiently large to robustly investigate whether VMS are present in metal-rich environments, and characterize properties of all YMCs. Targets were selected according to photospheric ages (1 to 10 Myr) supplemented by NIRCам archival observations (strong Pa alpha and/or location in tidal tail of NGC 4039). The Antennae is host to an exceptionally rich population of YMCs (some exceeding 10^6 Msun), such that the IMF is fully sampled. The youngest clusters are still embedded in their birth environment, and

are expected to each host 10^2 to 10^3 VMS. If VMS are present in high metallicity environments, NIRSpec spectroscopy will permit the detection of broad HeII emission, as well as nebular HI and HeI emission, plus PAH emission associated with photodissociation regions. Older clusters with ages of ~ 5 Myr will show broad emission from classical WR stars (without PAH's), with CO bandhead absorption and neutral metals from cool supergiants, seen in clusters with ages of ~ 10 Myr, permitting heavy elemental abundance determinations.

OBSERVING DESCRIPTION

NIRSpec/MSA is uniquely capable of obtaining spatially resolved 1-5 micron spectroscopy of multiple embedded young massive clusters (YMCs) in the Antennae. A master list of YMCs in the Antennae was drawn from Whitmore et al. (2010), supplemented by Bastian et al. (2009) plus clusters identified within regions of extended Pa alpha emission and in the tidal tail of NGC 4039 (from archival NIRCам/F187N imaging from GO2581). Complete NIR spectral coverage (aside from detector gaps) is required to fully characterize ages of clusters: G140H for broad HeII 1.01 micron emission, G235H for Pa alpha and CO 2.3 micron bandhead absorption, G395H for Br alpha, 3.3 micron PAH emission and potentially broad HeII 3.09 micron emission.

Four MSA configurations at PA=81 degrees permit a sufficient sample size (28 YMCs) to be observed to address our primary goals, using nodes in each 3 shutter slitlet, with 35 groups/integration and 1 integrations/exposure (NRSIRS2RAPID readout pattern), resulting in a continuum S/N of 50 with G140H + G235H + G395H gratings for typical clusters with $m(F160W/NICMOS)=18$ mag (corresponding to 60 micro Jy). High resolution gratings are required to distinguish broad HeII stellar emission (FWHM of 2500 km/s) from VMS and WR stars from nebular emission. These are designed according to 'midpoint' source centering constraints, with spare MSA apertures used to construct a global background. PA = 81 provides a higher fraction of high priority targets than PA = 242 (alternate visibility window). The brightest clusters could be observed at 1 micron from the ground with 8-10m class telescopes, but not in regions where transmission is poor (e.g. Pa alpha) or in the thermal IR (e.g. PAHs at 3.3 microns).

Proposal 10562 - Targets - Spectroscopy properties of young massive clusters in the Antennae: How common are very massive stars ...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(7)	YMC_Antennae	RA: 12 01 53.1680 (180.4715333d) Dec: -18 52 20.06 (-18.87224d) Equinox: J2000		
Comments: Description=[]					

Proposal 10562 - Observation 9 - Spectroscopy properties of young massive clusters in the Antennae: How common are very massive...

Observation	Proposal 10562, Observation 9: PA81-4configs-G140H-G235H-G395H-midpoint										Fri Mar 13 23:00:57 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(PA81-4configs-G140H-G235H-G395H-midpoint (Obs 9)) Warning (Form): Config c4 (#10) has 1 primary slits affected by failed closed shutters. (PA81-4configs-G140H-G235H-G395H-midpoint (Obs 9)) Warning (Form): Config c4 (#11) has 1 primary slits affected by failed closed shutters. (PA81-4configs-G140H-G235H-G395H-midpoint (Obs 9)) Warning (Form): Config c4 (#12) has 1 primary slits affected by failed closed shutters. (Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Visit 9:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
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	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	YMC_Antennae (56 sources)		jwst-nirspec-hr	1.5			
Reference Stars											

Proposal 10562 - Observation 9 - Spectroscopy properties of young massive clusters in the Antennae: How common are very massive...

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1	3 Shutter Slitlet	180.49860275 Degrees - 18.894445277777 777 Degrees	80.991230244591 85			3	3	1575.6
	2	2 (G235H/F170LP)	c1	3 Shutter Slitlet	180.49860275 Degrees - 18.894445277777 777 Degrees	80.991230244591 85			3	3	1575.6
	3	3 (G395H/F290LP)	c1	3 Shutter Slitlet	180.49860275 Degrees - 18.894445277777 777 Degrees	80.991230244591 85			3	3	1575.6
	4	1 (G140H/F100LP)	c2	3 Shutter Slitlet	180.49548387500 002 Degrees - 18.886475277777 777 Degrees	80.992250411399 17			3	3	1575.6
	5	2 (G235H/F170LP)	c2	3 Shutter Slitlet	180.49548387500 002 Degrees - 18.886475277777 777 Degrees	80.992250411399 17			3	3	1575.6
	6	3 (G395H/F290LP)	c2	3 Shutter Slitlet	180.49548387500 002 Degrees - 18.886475277777 777 Degrees	80.992250411399 17			3	3	1575.6
	7	1 (G140H/F100LP)	c3	3 Shutter Slitlet	180.45923770833 332 Degrees - 18.875032777777 79 Degrees	81.003958933946 42			3	3	1575.6
	8	2 (G235H/F170LP)	c3	3 Shutter Slitlet	180.45923770833 332 Degrees - 18.875032777777 79 Degrees	81.003958933946 42			3	3	1575.6
	9	3 (G395H/F290LP)	c3	3 Shutter Slitlet	180.45923770833 332 Degrees - 18.875032777777 79 Degrees	81.003958933946 42			3	3	1575.6
	10	1 (G140H/F100LP)	c4	3 Shutter Slitlet	180.48875287500 002 Degrees - 18.886218333333 318 Degrees	80.994422400202 13			3	3	1575.6
	11	2 (G235H/F170LP)	c4	3 Shutter Slitlet	180.48875287500 002 Degrees - 18.886218333333 318 Degrees	80.994422400202 13			3	3	1575.6
	12	3 (G395H/F290LP)	c4	3 Shutter Slitlet	180.48875287500 002 Degrees - 18.886218333333 318 Degrees	80.994422400202 13			3	3	1575.6

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Special Requirements	Group Visits within 53.0 Days Aperture PA Range 80 to 82 Degrees (V3 301.4254303 to 303.4254303) Visits Same PA MSA Planned Aperture PA 81.0000 to 81.0000 Degrees (V3 302.4254303 to 302.4254303)
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